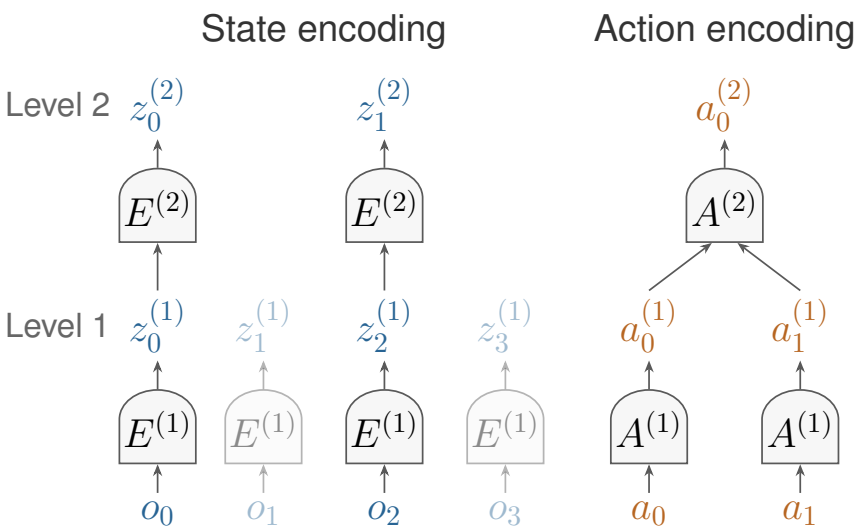
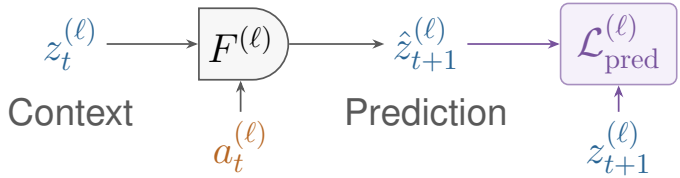
(a) Build hierarchy ($w_2 = 1$, $s_2 = 2$)



(b) Train each level

For each level $\ell = 1, 2$



Regularize encoded states

$$\text{SIGReg}\left(Z^{(\ell)}\right), \qquad Z^{(\ell)} = \{z_t^{(\ell)}\}_{\text{batch, time}}$$

Per-level objective

$$\mathcal{L}^{(\ell)} = \mathcal{L}_{\text{pred}}^{(\ell)} + \lambda_\ell \, \text{SIGReg}\left(Z^{(\ell)}\right)$$